

Robust Verification of Concurrent Stochastic Games

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- 1 Motivation
- 2 Preliminaries
- 3 Model: Robust/Interval CSGs (R/ICSGs)
- 4 Main Results
 - Key technique: Adversarial expansion
- 5 Algorithm
- 6 Experimental Results
- 7 Conclusion and Future Work

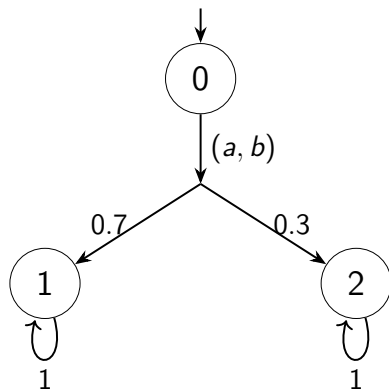
- Many multi-agent systems are *concurrent*, *stochastic*, and have *uncertain* transition probabilities.
- Standard CSG verification assumes exact transitions — not realistic.
- Transition probabilities are often:
 - learned from data
 - abstracted
 - noisy or incomplete

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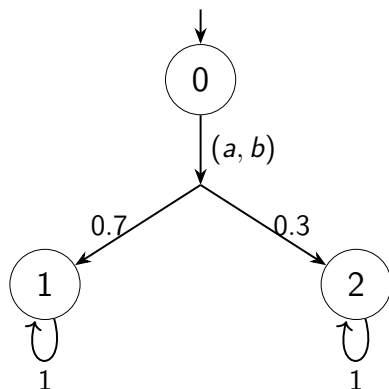


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Typical applications:

- autonomous coordination
- network protocols
- competitive resource allocation



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Assumption: transition probabilities are known exactly

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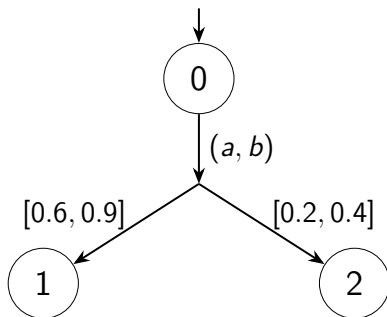
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Goal: **robust verification** (and synthesis) of CSGs under uncertainty.

Our Model: *Robust* CSG (RCSG)

- Uncertain transition function
- **Interval** uncertainty: each transition probability lies in a known interval

$$P(s' \mid s, a) \in [\underline{p}, \bar{p}]$$



- **Zero-sum:** nature adversarial against one player (minimises their value)
 - *robust* value, *robust* optimal strategies
- **Nonzero-sum:** nature minimises total utility
 - ε -*robust* NE (ε -RNE): ε -NE under *every* admissible model
 - *robust* SWNE (RSWNE): RNE that maximises *robust* (worst-case) social welfare

Robust Bellman equation

- **Zero-sum:** nature adversarial against one player (minimises their value)

$$V(s) = \sup_{\sigma_1 \in \mathcal{D}(A_1(s))} \inf_{\sigma_2 \in \mathcal{D}(A_2(s))} \left\{ r(s, \sigma) + \sum_{a \in A(s)} \sigma_{sa} \inf_{P_{sa} \in \mathcal{P}_{sa}} \sum_{s' \in S} P_{sas'} \cdot V(s') \right\}$$

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$$V_+(s) = \sup_{\sigma \in \Sigma_{\epsilon\text{-RNE}}} \left[r_+(s, \sigma) + \sum_{a \in A} \sigma_{sa} \inf_{P_{sa} \in \mathcal{P}_{sa}} \left\{ \sum_{s' \in S} P_{sas'} V_+(s') \right\} \right]$$

where $c_+ = c_1 + c_2$.

Key idea: Adversarial Expansion

- Model *nature* (uncertainty) as player 3 who chooses transitions *adversarially*
 - Zero-sum: nature aligns with opponent $\Rightarrow$ 2-player CSG
 - Nonzero-sum: nature minimises social welfare $\Rightarrow$ 3-player CSG + NE filtering
- Robust reasoning $\Rightarrow$ standard game solving

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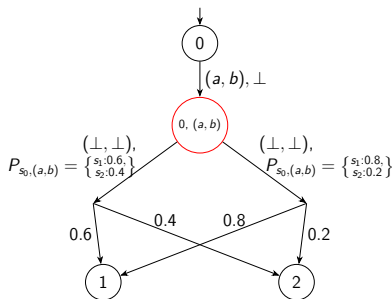
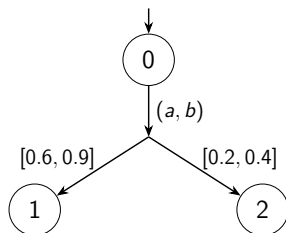
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 - allows *robust dynamic programming*
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 - allows *robust dynamic programming*
 - i.e., fix nature's choice of P first, then solve induced CSG
- Algorithm to compute robust value of the ICSG by reusing CSG algorithms (**without building $\mathcal{G}^{\mathcal{A}}$ explicitly!**)

RVI/RBI update for state $s \in S$ in $\mathcal{G} = (N, S, \bar{s}, A, \Delta, \check{P}, \hat{P})$

- 1: **for all** $a \in A(s)$ **do**
- 2: $P_{sa}^* \leftarrow \text{SOLVEINNERPROBLEM}(s, a, V_{prev}, \check{P}, \hat{P})$
- 3: **end for**
- 4: $\mathcal{Z} \leftarrow \text{CONSTRUCTNFG}(P_s^*, V_{prev})$
- 5: $V_{next}[s] \leftarrow \text{SOLVENFG}(\mathcal{Z})$



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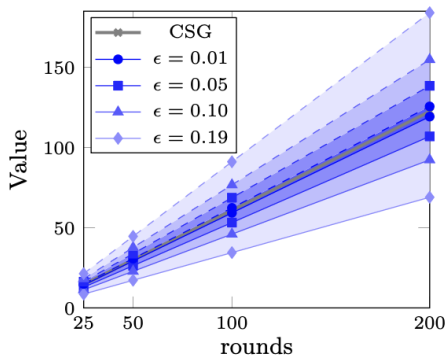


Figure 1: ICSG values over game size in the *Intrusion Detection* case study.

- Introduced robust concurrent stochastic games (RCSGs), with focus on the ICSG subclass.
- New equilibrium notions: RNE, RSWNE
- Key technique: *Value-preserving* reductions to standard CSGs
 - Zero-sum: ICSG $\rightarrow$ 2-player CSG
 - Nonzero-sum: ICSG $\rightarrow$ 3-player CSG + NE filtering
- Important result: Value-equivalence regardless of whether players or nature moves first
- Enabled algorithms for zero-sum and nonzero-sum ICSGs
 - Implementation in PRISM-games
 - Scales to millions of states

- Extend beyond interval uncertainty (e.g., weaker rectangularity).
- Broader objectives (LTL / richer temporal properties).
- Integration with learning pipelines to provide robust guarantees for learned policies.

Questions?